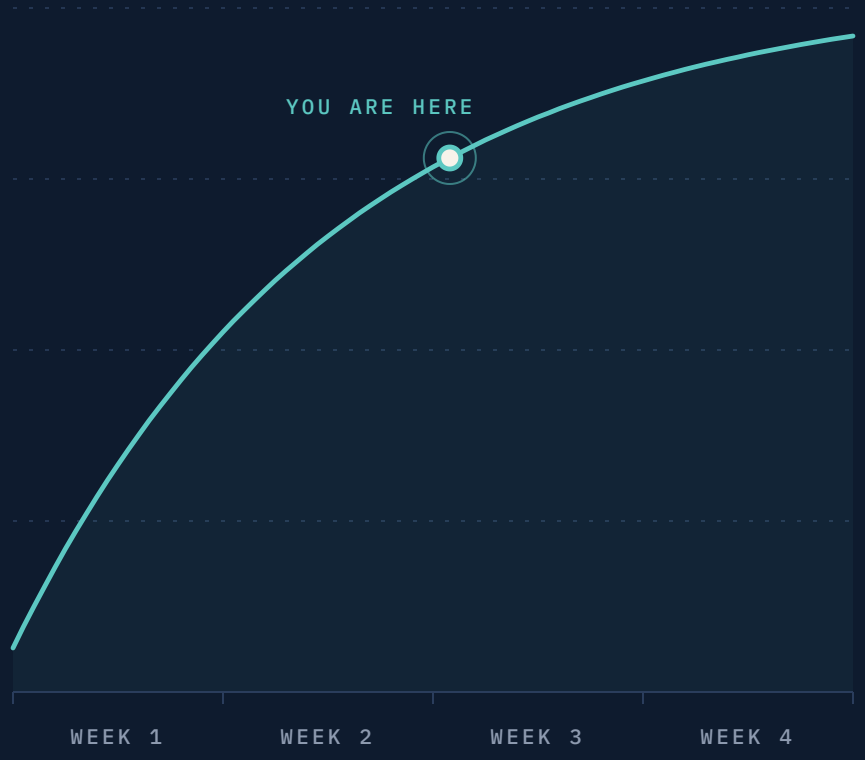
WEEK 2 · DAY 4 · OVERVIEW

Case Study

A First Look at the Data



Created by **Jeff Lenamon**

ABOUT 50 MIN WITH EXERCISES / 6 SECTIONS / 7 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Definition

Yesterday we checked the joins; nobody looked at each file on its own. The three systems now send these files every month, and nobody tells us what is wrong with them. Before tomorrow's report, tell me what each file holds, whether anything is missing or doubled, and that the face still ties to our 489,000,000. Give me the bonds to watch: below par, long-dated, small. Will the three join without losing or doubling a bond? And write the checks so they run themselves next month.

ABOUT 50 MIN WITH EXERCISES / 6 SECTIONS / 7 EXERCISES

Your Route Through Day 4

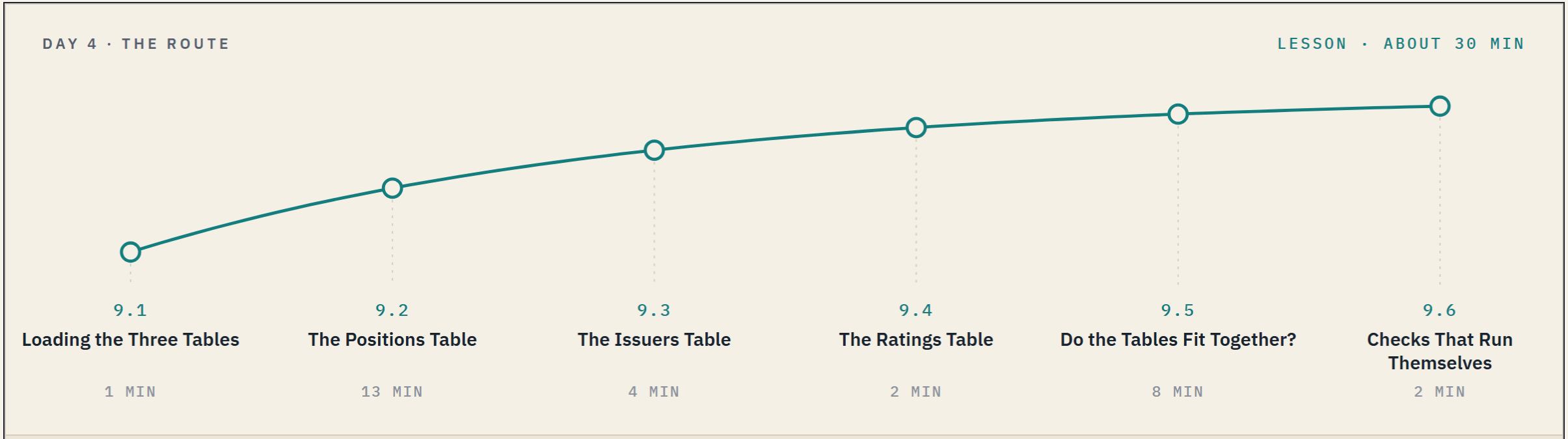


FIGURE 2 Six sections, in order, with about how long each takes.

Time. About 30 minutes for the lesson, then about 21 for the seven exercises.

9.1 Loading the Three Tables

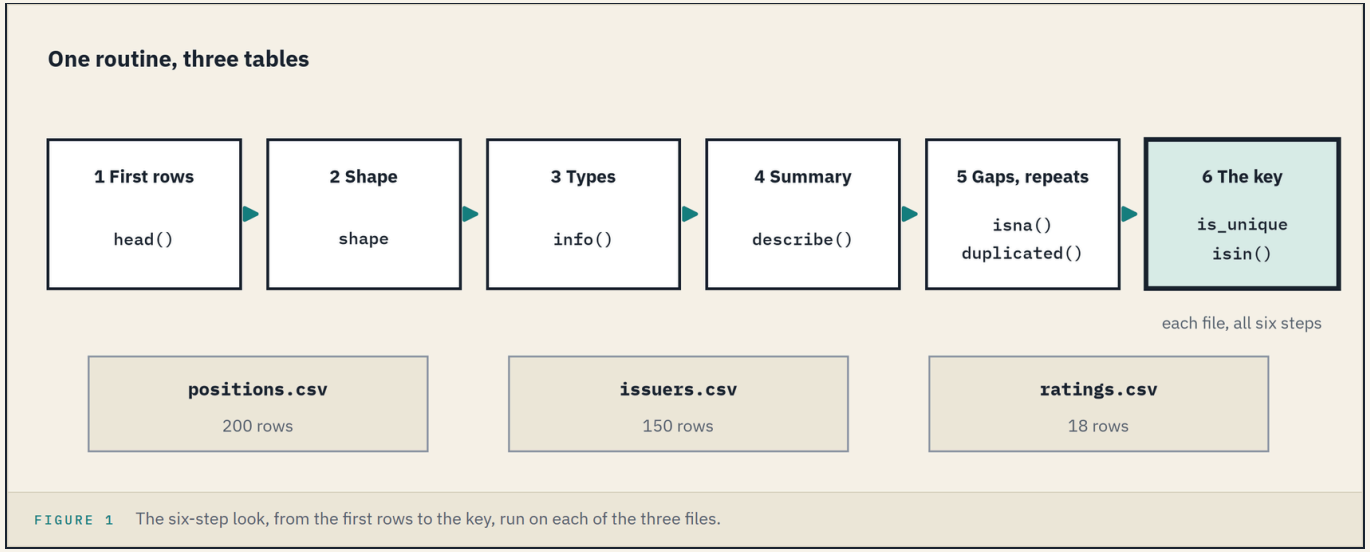


FIGURE 1 The six-step look, from the first rows to the key, run on each of the three files.

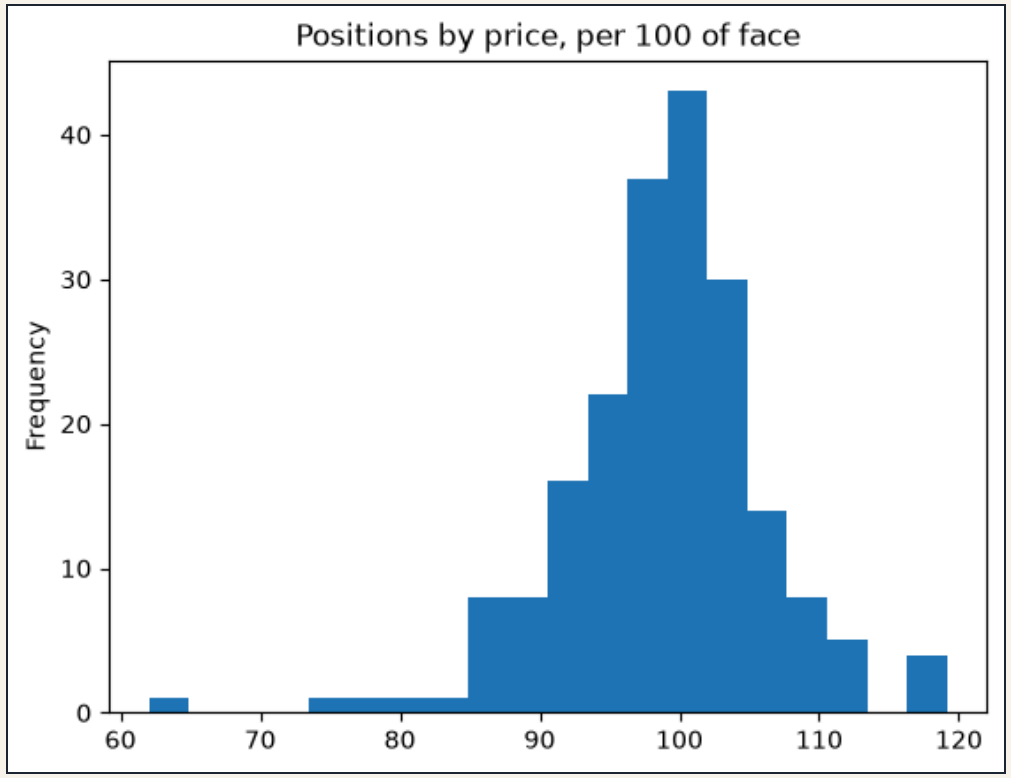
LOAD THE THREE FILES

```
print(positions.shape, issuers.shape, ratings.shape)
```

```
(200, 6) (150, 5) (18, 4)
```

Three sizes, so a row stands for something different in each table. Find out what.

9.2 The Positions Table



THE SUMMARY

	COUPON	PRICE	PAR_HELD
count	200.000000	200.000000	2.000000e+02
mean	5.937500	98.828295	2.445000e+06
std	1.648435	7.404931	1.366818e+06
min	3.250000	62.045000	2.500000e+05

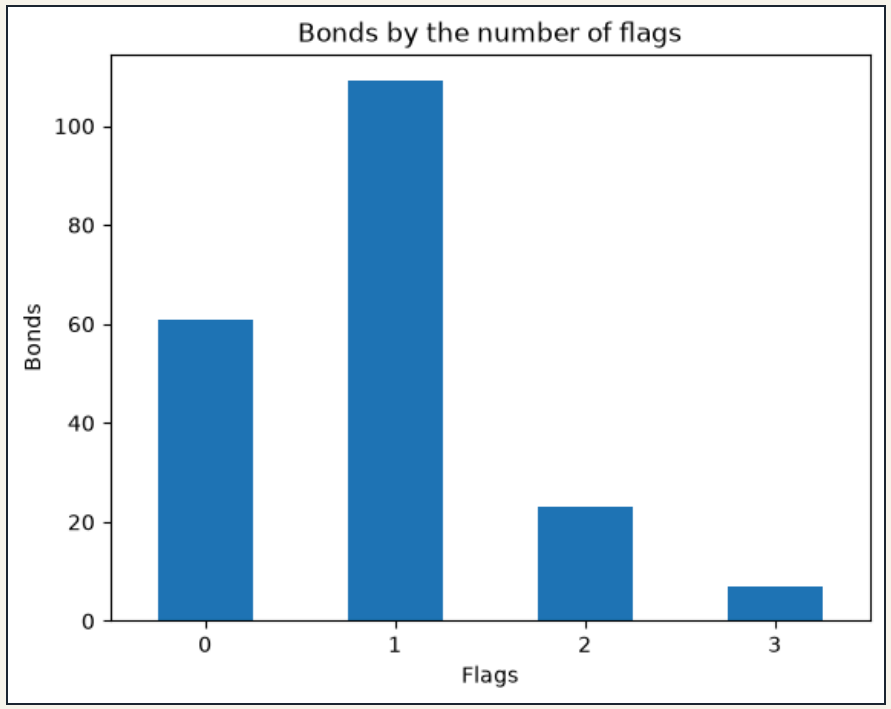
THE LOWEST PRICES

```
positions.nsmallest(3, 'price')
```

	cusip	ticker	coupon	maturity	price	par_held
17	99009JAA9	CLDB	9.500	2034-12-06	62.045	3000000
117	99054CAG5	NEVX	3.875	2048-01-06	73.873	1000000
150	99053BAB9	SYLX	3.625	2055-09-03	76.903	4500000

One price sits far below the rest: CLDB. The files show it; they cannot say why, so the analyst asks the desk.

Bonds to Watch



THREE FLAGS

```
flags = pd.DataFrame({
    'below_par': positions['price'] < 100,
    'long_dated': maturity_dates > pd.Timestamp('2040-12-31'),
    'small': positions['par_held'] < 1000000,
flags.sum()
```

```
below_par     113
long_dated     34
small          29
```

EXCEL SIDE BY SIDE

JOB	PANDAS	EXCEL
Flags	`positions['price'] < 100`, `flags.sum(axis=1)`, `flags.sum()`	`=IF(E2<100, 1, 0)` in G, `=SUM(G2:I2)`, `=SUM(G2:G201)`

Seven bonds trip all three tests. The list describes the file; it judges no bond.

9.3 The Issuers Table

TYPES AND GAPS

```
issuers.info()
print('Repeated rows:', issuers.duplicated().sum())
```

```
 0   ticker   150 non-null    str
 1   issuer   150 non-null    str
 2   sector   150 non-null    str
 3   country  150 non-null    str
 4   rating   150 non-null    str
Repeated rows: 0
```

ISSUERS PER SECTOR

```
print(issuers['sector'].value_counts())
print(issuers['country'].unique().tolist())
```

```
sector
Materials          18
Utilities          18
Energy             17
```

AN AVERAGE RATING?

```
issuers['rating'].mean()
```

```
TypeError: Cannot perform reduction 'mean' with string dtype
```

A rating is letters. Its number, the score, is in the third table.

9.4 The Ratings Table

THE SCALE

	RATING	SCORE	BUCKET	GRADE
0	AAA	1	AAA	Investment Grade
1	AA+	2	AA	Investment Grade
2	AA	3	AA	Investment Grade
3	AA-	4	AA	Investment Grade
4	A+	5	A	Investment Grade
5	A	6	A	Investment Grade
6	A-	7	A	Investment Grade
7	BBB+	8	BBB	Investment Grade
8	BBB	9	BBB	Investment Grade
9	BBB-	10	BBB	Investment Grade

EVERY SCORE ONCE

```
print('Scores 1 to 18:', ratings['score'].tolist() == list(range(1, 19)))
```

```
Scores 1 to 18: True
```

TWO GRADES

```
ratings.groupby('grade')['score'].agg(['count', 'min', 'max'])
```

```
                 count  min  max
grade
High Yield           8   11   18
Investment Grade    10    1   10
```

18 notches, two grades that neither overlap nor leave a gap, nothing missing: a complete scale.

9.5 Do the Tables Fit Together?

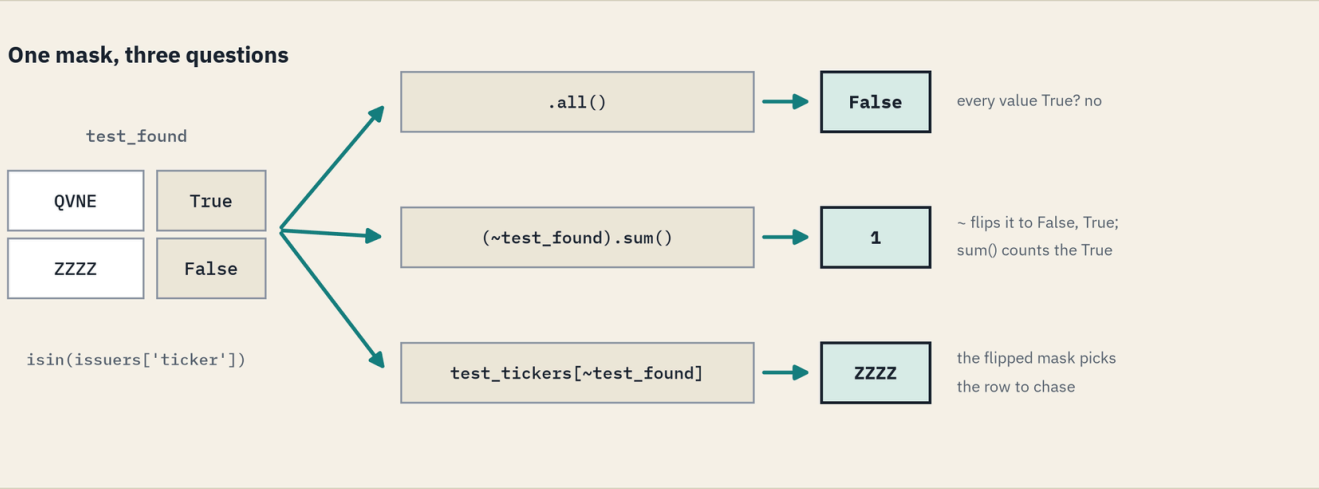


FIGURE 4 The mask True, False read three ways: all gives False, the tilde flips it, and the flipped mask picks ZZZZ.

```
print('cusip in positions:', positions['cusip'].is_unique)
print('ticker in issuers:', issuers['ticker'].is_unique)
print('rating in ratings:', ratings['rating'].is_unique)
print('ticker in positions:', positions['ticker'].is_unique)
```

```
cusip in positions: True
ticker in issuers: True
rating in ratings: True
ticker in positions: False
```

Many on one side, one on the other: the shape Combining Part 1's `validate='many_to_one'` demands.

9.6 Checks That Run Themselves

SIX CHECKS

```
assert positions.isna().sum().sum() == 0, 'the positions have missing values'
assert positions.duplicated().sum() == 0, 'the positions have repeated rows'
assert issuers['ticker'].is_unique, 'a ticker appears twice in the issuers table'
assert ratings['rating'].is_unique, 'a rating appears twice on the scale'
assert positions['ticker'].isin(issuers['ticker']).all(), 'a ticker is missing from issuers'
assert issuers['rating'].isin(ratings['rating']).all(), 'a rating is not on the scale'
print('Six checks passed:', positions.shape, issuers.shape, ratings.shape)
```

```
Six checks passed: (200, 6) (150, 5) (18, 4)
```

A CHECK THAT FAILS

```
assert positions['ticker'].is_unique, 'the ticker repeats, so it cannot identify a bond'
```

```
AssertionError: the ticker repeats, so it cannot identify a bond
```

The findings now run on next month's files, and stop at the first one that fails.

You Can Now

- Give any new table the same six-step look, and say what one row stands for.
- Convert date text to dates on the side, leaving the table as it arrived.
- Count gaps and repeated rows with **`isna()`** and **`duplicated()`**.
- Count flags down a column, and across a row with **`axis=1`**.
- Test a key with **`is_unique`** and **`isin()`**, and read a mask with **`.all()`** and **`~`**.
- Write down what a join should return, and the checks it depends on as **`assert`** lines.

NEXT
WEEK 2 · DAY 5

Case Study: Joining the Tables

The three tables checked yesterday are joined into one book, and the desk's questions are answered from it.

